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\documentclass{article}
\usepackage{array}
\usepackage{geometry}
\geometry{margin=1in}
\begin{document}

\section*{Comparison of Language Modeling Approaches}

\renewcommand{\arraystretch}{1.5}
\setlength{\tabcolsep}{8pt}

\begin{tabular}{|>\raggedright m{2.5cm}|>\raggedright m{4cm}|>\raggedright m{3.5cm}|>\raggedright m{6cm}|>\raggedright m{3.5cm}|}
\hline
\textbf{Concept} & \textbf{Definition} & \textbf{Problem it Solves} & \textbf{How it Solves (Types, Formula, Example)} & \textbf{Limitation} \\
\hline

\textbf{N-gram LM} &
Statistical language model that predicts the next word based on the last $n$ words. &
&
Provides a simple way to estimate conditional probabilities of word sequences. &
Formula:  $P(w_t | w_{t-n+1}, \dots, w_{t-1})$ . \\
Example: Bigram model predicts ``sat'' after ``cat''. &
Suffers from sparsity (rare sequences), requires huge storage, cannot handle long-range dependencies. \\
\hline

\textbf{Perplexity} &
Evaluation metric for language models; measures how well a model predicts a sample. &
&
Quantifies predictive performance of a language model. &
Formula:  $PPP = 2^{\{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)\}}$ . \\
Example: Lower perplexity indicates better prediction accuracy. &
Only a numeric score; does not explain why errors occur or provide qualitative insight. \\
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\textbf{RNN} &
Neural network with recurrent connections that processes sequences word by word. &
Handles variable-length input and captures sequential dependencies. &
Formula:  $h_t = f(w_{x_t} + U_{h_{t-1}})$ . \\
Example: Predicting the next word in a sentence. &
Sequential processing (slow), vanishing/exploding gradients, struggles with long dependencies. \\
\hline

\textbf{LSTM} &
Advanced RNN with memory cell and gating mechanisms (input, forget, output gates). &
&
Addresses vanishing gradient problem and remembers long-term dependencies. &
Cell state update:  $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$ . \\
Example: Captures relation between ``book'' and ``was amazing''. &
Computationally expensive, still sequential in nature. \\
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\textbf{Seq2Seq} &
Encoder-Decoder RNN architecture mapping input sequence to output sequence of different length. &

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Enables variable-length outputs, e.g., machine translation. &
Encoder compresses input into a vector; Decoder generates output step by step. \newline
Example: English sentence $\rightarrow$ French translation. &
Information bottleneck: entire input compressed into one vector, loses detail for long sentences. \\

Attention &
Mechanism that allows decoder to focus on relevant encoder states instead of a single vector. &
Solves Seq2Seq bottleneck, improves handling of long-range dependencies. &
Formula: $\text{Attention}(q, k, v) = \text{softmax}(qk^T)v$. \newline
Types: Additive (Bahdanau), Dot-product (Luong), Self-attention, Multi-head. \newline
Example: Decoder attends to ``yesterday'' when generating ``hier''. &
Adds complexity; self-attention has quadratic cost in memory and compute. \\

\end{tabular}

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